

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2015

BIOLOGY
Higher 2

Paper 3

24th AUG 2015/ Monday
2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

At the end of examination,

1. fasten all your work securely together

FOR EXAMINER'S USE	
1	/13
2	/13
3	/14
	/40
4	/12
5	/20
TOTAL	/72

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 15 printed pages and 1 blank page

Answer **all** questions

Question 1

Protein X is a single polypeptide consisting of 35 amino acids. It is produced in minute quantities in a particular species. The rate of transcription to synthesise the mRNA for Protein X is known to be constantly low.

A researcher has discovered that Protein X has the ability to reduce wrinkles instantly when applied topically on the skin. Realising its potential to make money, the researcher intends to produce large amounts of Protein X in a short period of time.

- (a) Suggest why Protein X is only produced in minute quantities in the particular species. [1]

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The genome sequence of the particular species has been determined. Thus, the sequence that codes for Protein X has been identified. The complete sequence of the non-template strand coding for the complete Protein X is shown in **Fig. 1.1** in the 5' to 3' orientation.

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1    CTGCAGGATA TGGCTGTGCT GTACGTCAGT GGATCCGGCT GCATGCAAGA    50
51   AAGCGGCTGG ATCGATCGAA TTCTCGTGTC GTTGCTGCAC TTGCATGCTT    100
101  TGTCGAGCTC CCGTTGATGC CTGCAGCATT
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Fig. 1.1

- (b) Label and indicate with two circles on **Fig. 1.1** the first and last codon of the coding sequence of Protein X. [2]

To obtain the gene of interest, the researcher contemplated between using the genomic sequence in **Fig. 1.1** and creating a cDNA library to obtain a cDNA sequence of Protein X.

- (c) Explain what is meant by a cDNA library. [3]

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The researcher concluded that there was no difference between using the genomic sequence in **Fig. 1.1** and the cDNA sequence of Protein X to produce the functional protein.

(d) Suggest the basis of the researcher's conclusion. [1]

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The researcher decided to use the genomic DNA and the plasmid pUC18 to clone the gene.

He proceeded to isolate the genomic DNA from the particular species. Using this genomic DNA, the researcher then carried out Polymerase Chain Reaction (PCR) to obtain sufficient amount of the DNA containing the sequence shown in **Fig. 1.1**.

Fig. 1.2 shows the diagram of the plasmid pUC18 and the positions of the restriction sites found in the multiple cloning site (MCS).

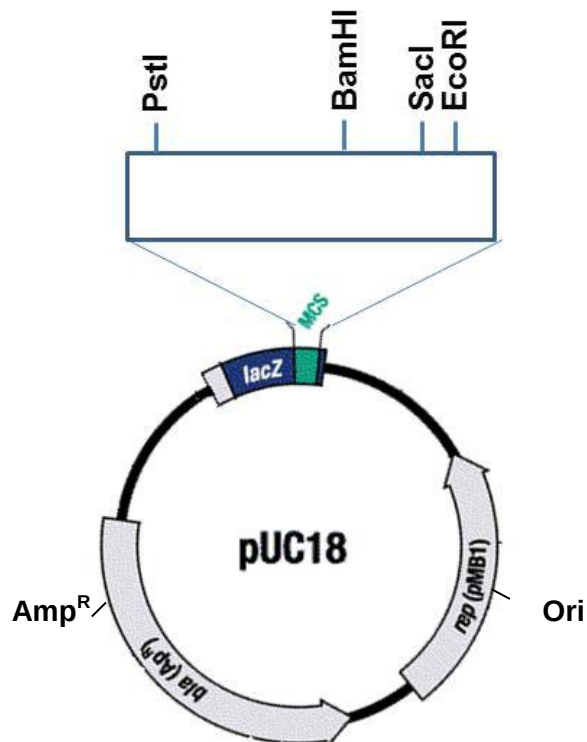


Fig. 1.2

(e) **Table 1.3** shows the restriction sites of four restriction enzymes.

name	site
<i>Bam</i> H1	↓ 5'... GGATCC...3' 3'... CCTAGG...5' ↑
<i>Eco</i> RI	↓ 5'... GAATTC...3' 3'... CTTAAG...5' ↑
<i>Pst</i> I	↓ 5'... CTGCAG...3' 3'... GACGTC...5' ↑
<i>Sac</i> I	↓ 5'... GAGCTC...3' 3'... CTCGAG...5' ↑

Table 1.3

- (i) With reference to **Fig. 1.1**, **Fig 1.2** and **Table 1.3**, state the restriction enzyme that the researcher should use to produce the recombinant plasmid. [1]

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- (ii) Describe the steps that the researcher would carry out in the formation of the recombinant plasmid using the restriction enzyme in (i). [4]

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The recombinant plasmids were then inserted into bacterial cells. However, not all transformed cells with the recombinant plasmid were able to produce functional Protein X.

- (iii) Suggest why not all bacteria that have taken up the recombinant plasmid would produce a functional Protein X. [1]

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[Total: 13]

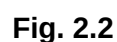
Huntington's disease (HD) is a lethal neurodegenerative disorder that is inherited as an autosomal dominant condition. Due to the late onset of symptoms, patients with HD may unknowingly pass on the mutant HD allele to their children. Symptoms in individuals as well as confirmation of diagnosis in other family members are used to determine the diagnosis.

Diagram illustrating the RFLP analysis for the G8 probe. The probe is shown as a black bar. The DNA sequence is represented by a horizontal line. The restriction sites are indicated by arrows pointing down to the line. The fragment sizes (in kb) are labeled above the line.

RFLP Variant 1: The DNA sequence shows restriction sites at 15.0, 4.9, 2.3, and 8.4 kb. The HD allele is indicated.

RFLP Variant 2: The DNA sequence shows restriction sites at 15.0, 3.7, 1.2, 2.3, and 8.4 kb. The HD allele is indicated.

Fig. 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.



(a) With reference to **Fig. 2.1** and **2.2**,

- (i)** outline the process of nucleic acid hybridization that allows the RFLP pattern of the 30 individuals to be visualized. [4]

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- (ii)** identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

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- (iii)** suggest one possible reason for the difference in the DNA band patterns among unaffected individuals. [2]

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- (b) Other applications of RFLP are in the areas of genetic fingerprinting and in genomic mapping.

Explain why genes especially protein-encoding genes are seldom used as markers for genetic fingerprinting. [1]

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- (c) Explain how PCR facilitates RFLP analysis. [2]

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- (d) The human genome project holds benefits for many fields, from molecular medicine to human evolution.

Importantly, scientists are now able to test for many inherited diseases.

State **one** advantage and **one** disadvantage of the ability to test for inherited diseases. [2]

Advantage:

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Disadvantage:

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[Total: 13]

Question 3

Plant tissue culture has many applications. These include commercial production of genetically modified plants and the culturing of plant cells in a bio-reactor for the production of valuable products.

Early attempts to culture plant tissue on a culture medium were not always successful.

The culture medium contained a range of substances that were known to be requirements for plant cell growth and division. These included:

- elements, present as minerals ions
- a sugar suitable for plant cells to use

(a) Explain why a sugar was included in the plant tissue culture medium. [1]

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(b) Outline how plants are cloned by tissue culture. [4]

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(c) Explain why these plants are described as clones. [2]

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Investigations have shown that the original culture medium lacked key components called plant growth regulators needed for successful plant cell growth and division. Although early attempts to culture plant tissue were not always successful, some investigators have reported successful growth using root tips. Root tips are known to contain meristematic tissues.

- (d) Suggest one reason why plant tissue culture was more successful using organs such as root tips. [1]

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- (e) Outline **one** advantage and **one** disadvantage of using tissue culture technique to produce plants. [2]

Advantage:

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Disadvantage:

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- (f) Distinguish between zygotic stem cells and embryonic stem cells. [4]

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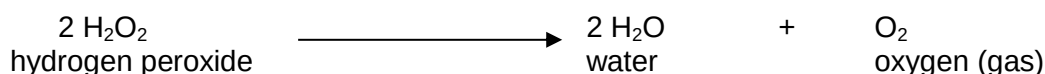
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[Total: 14]

Question 4

Potato catalase is found in potato tissue and will be present in extracts made from potatoes. The reaction catalysed by catalase is shown below:



The presence of inhibitors may affect enzymatic reactions. There are two main types of inhibitors, namely, competitive and non-competitive inhibitors. It is known that the presence of certain metal ions such as copper (Cu^{2+}) would result in the inhibition of the reaction catalysed by catalase.

You are required to plan, but not carry out, an investigation to determine if Cu^{2+} ions function as a competitive inhibitor or a non-competitive inhibitor of catalase.

The rate of reaction of the enzyme catalase can be measured by the rate of oxygen gas production. Oxygen can be collected by using a delivery tube from the reaction flask to a 20cm^3 gas syringe.

You must use:

- 20 cm^3 gas syringe
- 250 cm^3 conical flask with rubber bung
- Delivery tube
- Potato extract solution
- 20% hydrogen peroxide stock solution
- 5 mol dm^{-3} Copper (II) Sulphate (CuSO_4) solution
- Retort stand

You should select from the following apparatus:

- Normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Timer e.g stopwatch
- Syringes
- Bunsen burner with tripod, gauze and bench mat
- Thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams if necessary,
- identify the dependent and independent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment

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Question 5

Free-response question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

- (a) Describe cystic fibrosis as an example of a genetic disease and discuss its treatment using gene therapy. [9]
- (b) Discuss the ethical and social considerations of gene therapy. [5]
- (c) Using named examples, discuss how genetically modified organisms may solve the demand for food in the world. [6]

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